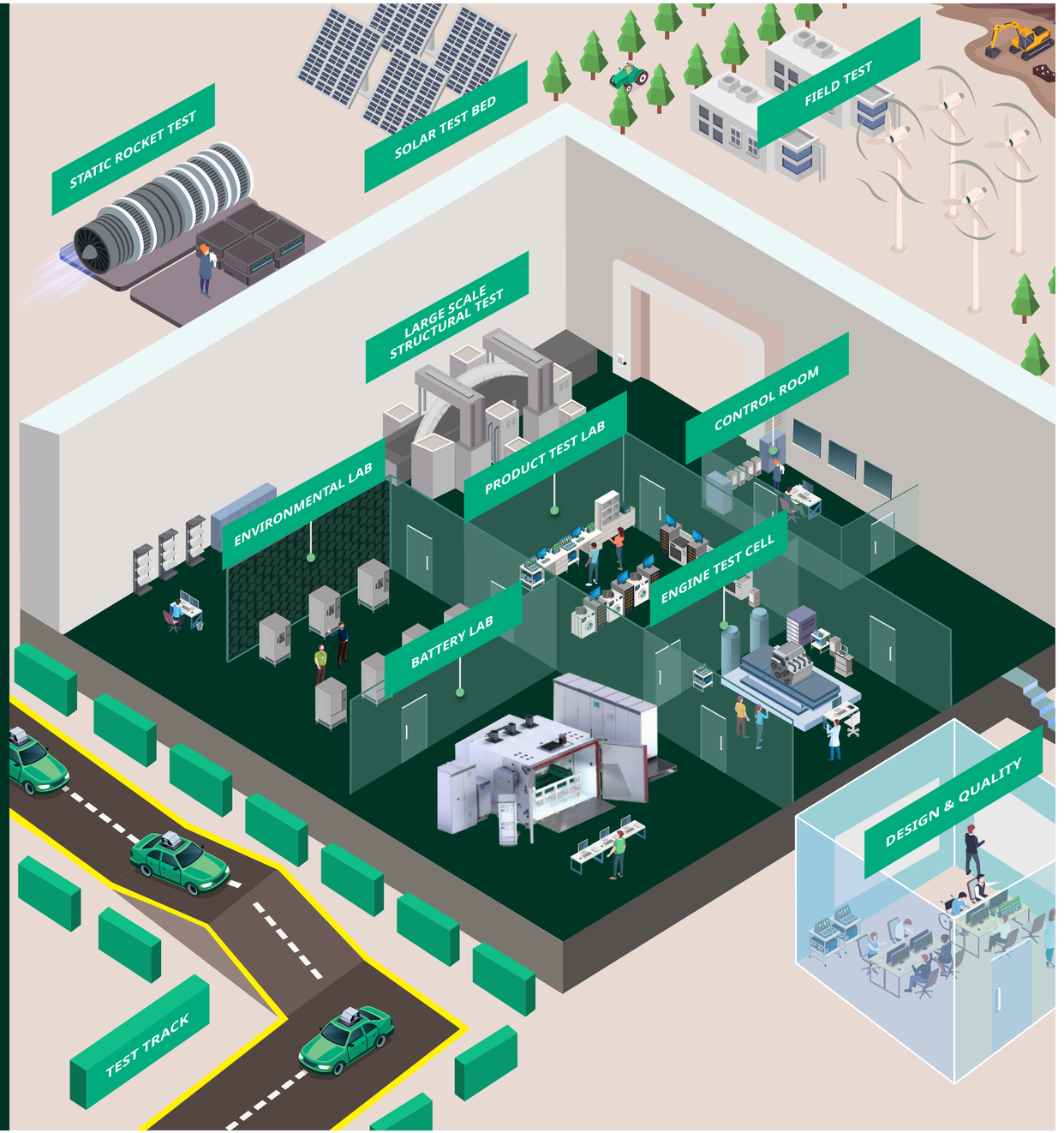
Twelve Tips to Level Up Your Mechanical Test Lab



FOREWORD

This document aggregates decades of experience from working with engineers in validation test labs and research testbeds across all industries. We've been to your labs. We've seen what you're doing. Some of you are setting the standard and want help staying on top. Others know they need help, but don't know where to start. This collection will help both.

Each consideration in this ebook could be its own full-length workshop. Use this guide to elevate the criteria important to you.

Finally, we're here to help. You can reach out to your NI account manager, the NI Discussion Forums, NI's authorized distributors, or the expansive NI Partner network of systems integrators and consultants.

Note

- There is no one right answer. The following are considerations to combine with your business operations and culture as you take your test program to the next level.
- These tips are not NI-specific—they are intended to be helpful regardless of the vendors, tools, or technologies you use.



TIP #1

Form a Committee

A modern lab is an efficient asset and a competitive differentiator for your business. To reach this goal, understand your stakeholders:

- Lab and systems owners with deliverables and deadlines
- Internal data customers with deadlines and an NPI process
- Business leaders with KPIs on executive dashboards
- Software developers
- Test system operators with tracking metrics
- Hardware technicians

For test programs with sizable budgets and executive-level interest, form a steering committee with the appropriate stakeholders to make informed decisions.

For test programs that are smaller and don't require a committee, consider documenting your stakeholders and interviewing each of them as you plan your development. What are their goals and how can your test system/lab help them achieve those goals?



TIP #2

Get IT Involved Early (Security)

If your test system is connected to anything beyond a local lab network, you should evaluate security needs and policies. Get IT involved early (even if just for consultation) to understand how your corporate security policies and whitelisted/blacklisted technologies will shape your deployed system. Here are some key recommendations:

01

Create a plan to manage software updates. Test system owners, security patch policies, and OS system patches will have different schedules. Consider an update prioritization matrix and schedule.

02

Document access rights based on user roles (operator, developer, maintenance, and so forth). Note that some IT security policies require access revocation within an allotted time window, so it's not just having access, it could be managing it as well.

03

Create a software bill of materials (SBOM) that lists your software technologies, packages, and vendors. Some industries have started to require SBOMs. Smaller businesses may need an SBOM as they seek funding. SBOMs are a hard trend—and chances are if you haven't been asked for an SBOM yet, you will in less than two years. RemindMe! – 2 years

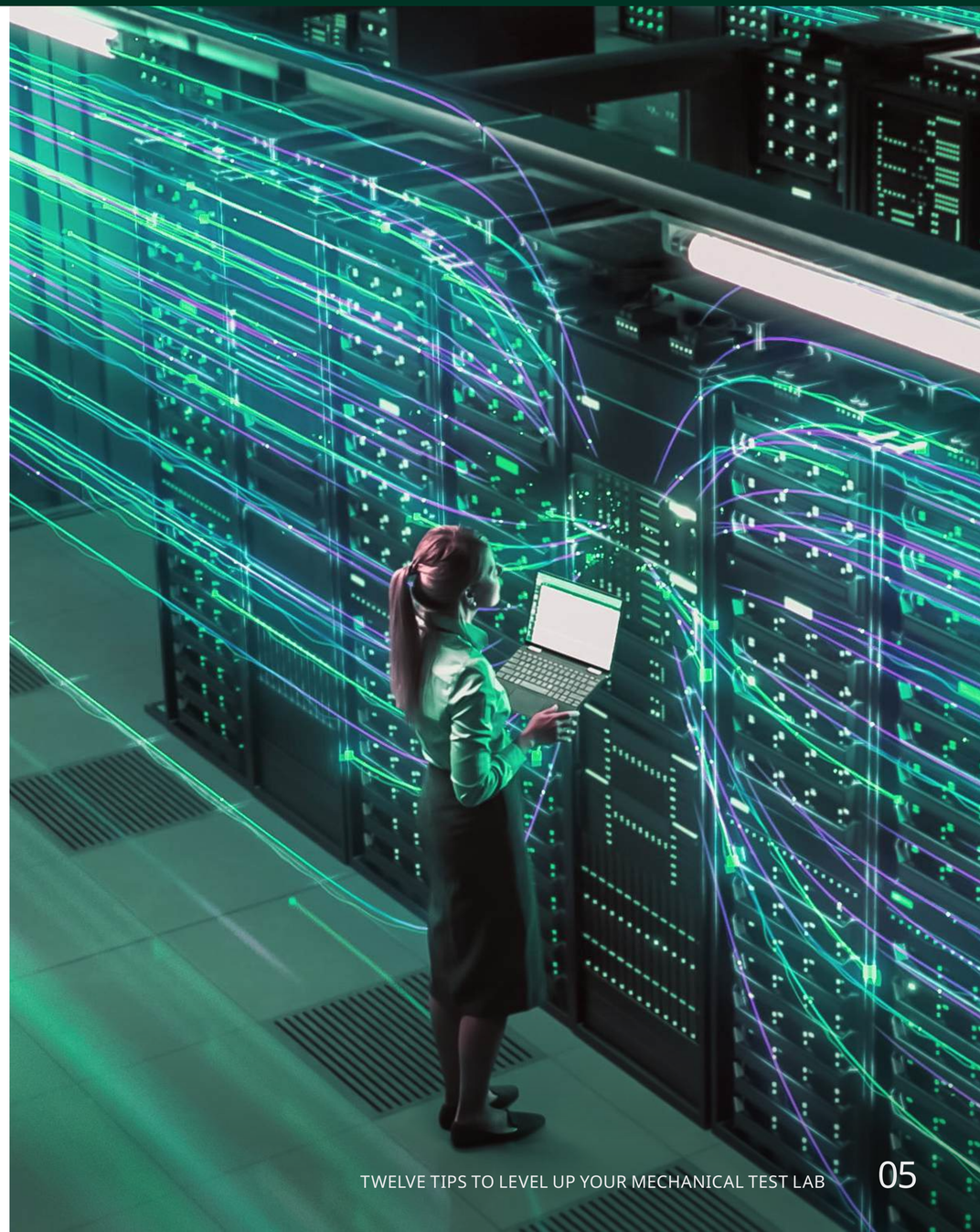


TIP #3

Have a Data Strategy

Whether you have multiple test systems for different mechanical components, or you're testing large assets like gas turbines, you're going to generate data. Modern labs aren't concerned with large data sets because they know data is a huge value-add and plan for it. You should, too. Consider:

- Storage locations including distributed on-premises (at each networked station), on-premises central server, or cloud.
- Data aging strategies. For example, how you treat raw data may differ from how you treat data reports and analysis. Aging strategies could end in data deletion, or they could end in lower-cost cold storage depending on business needs.
- Data security, backup, and encryption requirements. (This is another reason to get IT involved early.)
- Data access. Is there a set person or team that controls the data—or is your solution more open for the design team to self-serve? The situation could be different based on the type of data.
- Data analysis. Eighty percent of companies report having more data than they can efficiently mine for insight. Make sure you balance investments for analyzing data with investments for creating data. Keep in mind that automation goes a long way here!



TIP #4

Prepare the Data Around the Data

Channel Names

A good naming convention will be self-documenting and scalable. Consider measurement type and sensor location for each channel name and use an abstraction layer to map that to the physical channel on your instrumentation.

Metadata

Find out what data the design team, test system maintenance, or other internal customers need to know. Some examples include:

- Environmental conditions
- Product line
- Release status (prototype, build revision, and so forth)
- Firmware version

Missing contextual data can lead to uncertainty, design risk, and costly retests. Conversely, adding more data takes time and a more thoughtful software architecture.

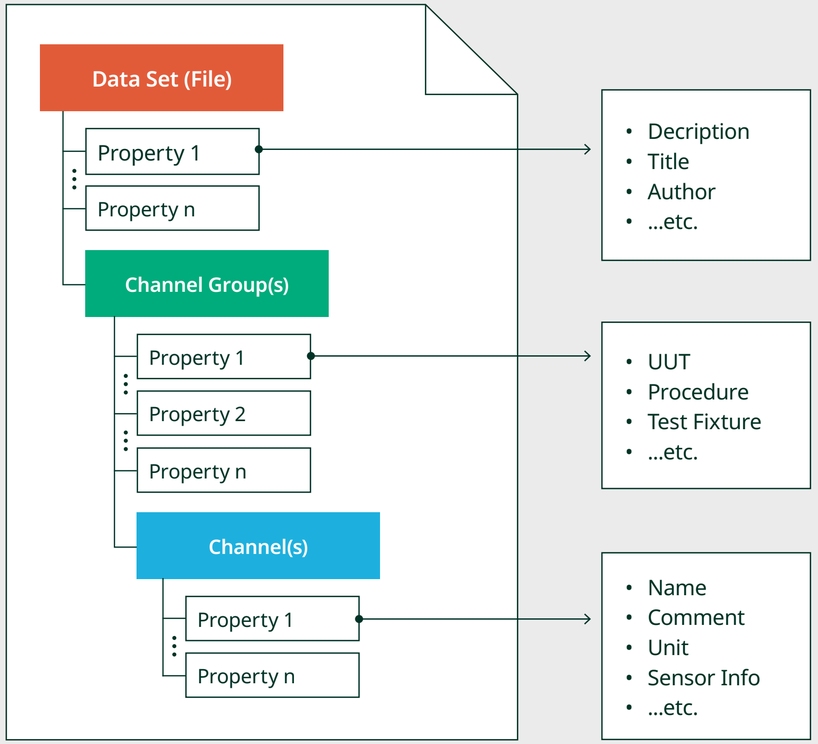
Consider:

01
Documenting your channel name schema as an internal standard

02
Interviewing your internal customers to better understand the context they need for the data

03
Including the ability to add more metadata in the future

04
Automating as much metadata logging as possible with software tools

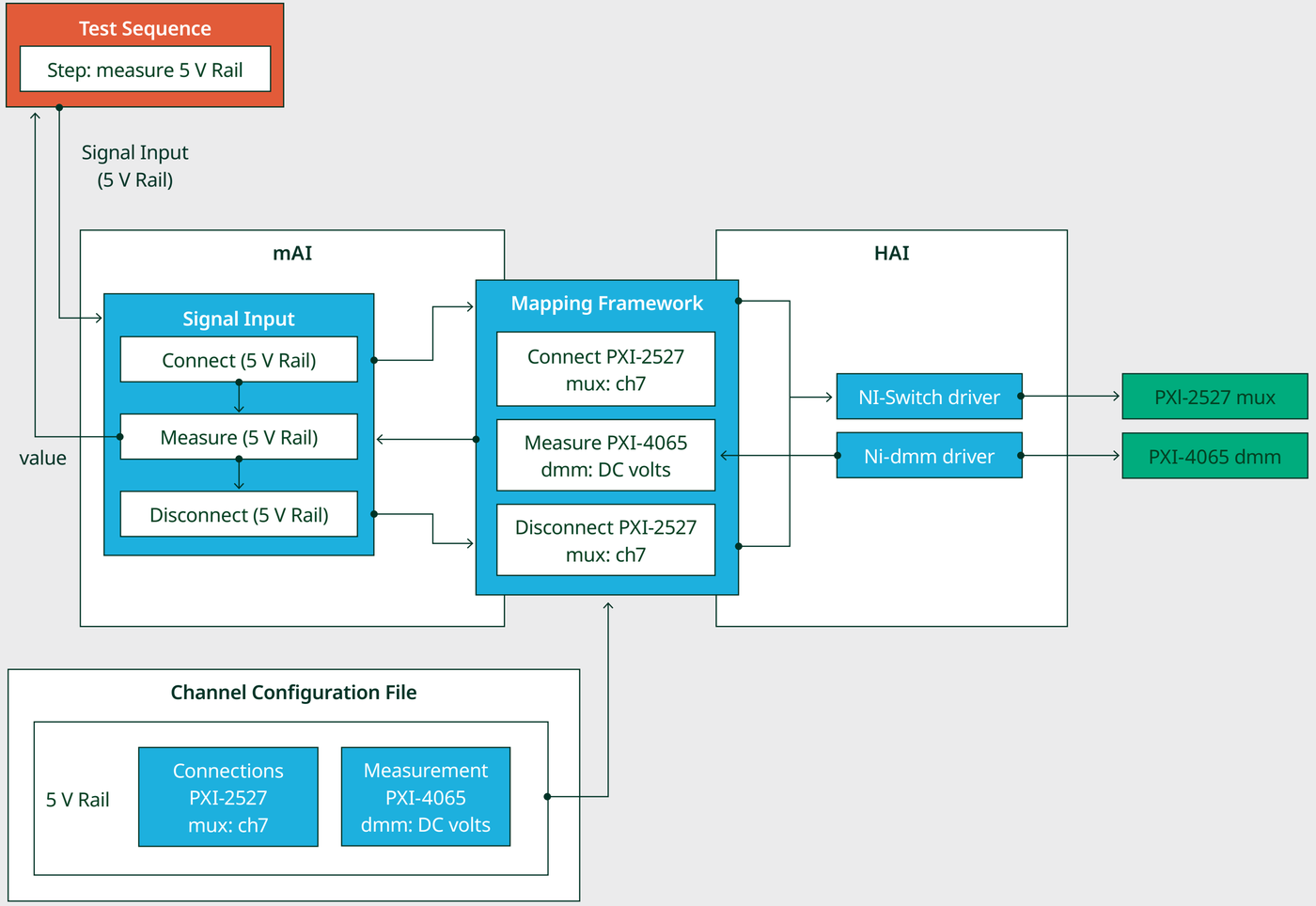


TIP #5

Design Systems with Abstraction Layers

Test system abstraction separates your test code from code that is specific to measurement equipment or technology stacks. For hardware, this is referred to as a hardware abstraction layer (HAL), and if implemented properly has several benefits:

- Encourages parallel development and modular code (which is the best practice)
- Supports multiple test bench setups with the same software easily
- Guards against hardware and software obsolescence since the tech-specific software calls are isolated from the rest of the codebase
- Protects from vendor “lock-in” and potential price increases



TIP #6

Be Flexible and Plan for Change

Change happens. It could come from technology obsolescence, market trends, government regulations and contracts, test team personnel changes, product features, or maybe from strategic leadership decisions. You need to prepare for the changes that will happen.

01

Develop your software on a modular architecture for a more sustainable system.

02

Use abstraction layers to make it easier to switch out different instrumentation and measurement equipment.

03

Modular systems can have a larger upfront cost but when paired with a hardware abstraction layer, save significant time and cost for future updates.

04

Your system may need to grow. Larger products like aircraft wings, rockets, bridges, wind turbine blades, steam turbines, and mining equipment can require thousands of channels. If this could happen to you, have a plan to scale to meet your needs such as network synchronization between hardware nodes or filling up chassis installed in 19" equipment racks.

05

Monitor your mix of open source, open platform, and proprietary technology. Proprietary is easier to develop, but harder to change. Open source is the opposite. Open platform is a hybrid. Ask “What do I do if XYZ vendor goes out of business?” and “How do I add new capabilities to this system?”



TIP #7

Make Automation Part of Your Culture

Automation saves people-time, reduces human error, helps maximize test system utilization, and increases test throughput. Here are some examples of how automation can help.

01

Send an SMS text to stakeholders when tests need attention or when complete to minimize downtime.

02

Automate data analysis, report generation, and transmission to data customers to reduce manual downloads and emails.

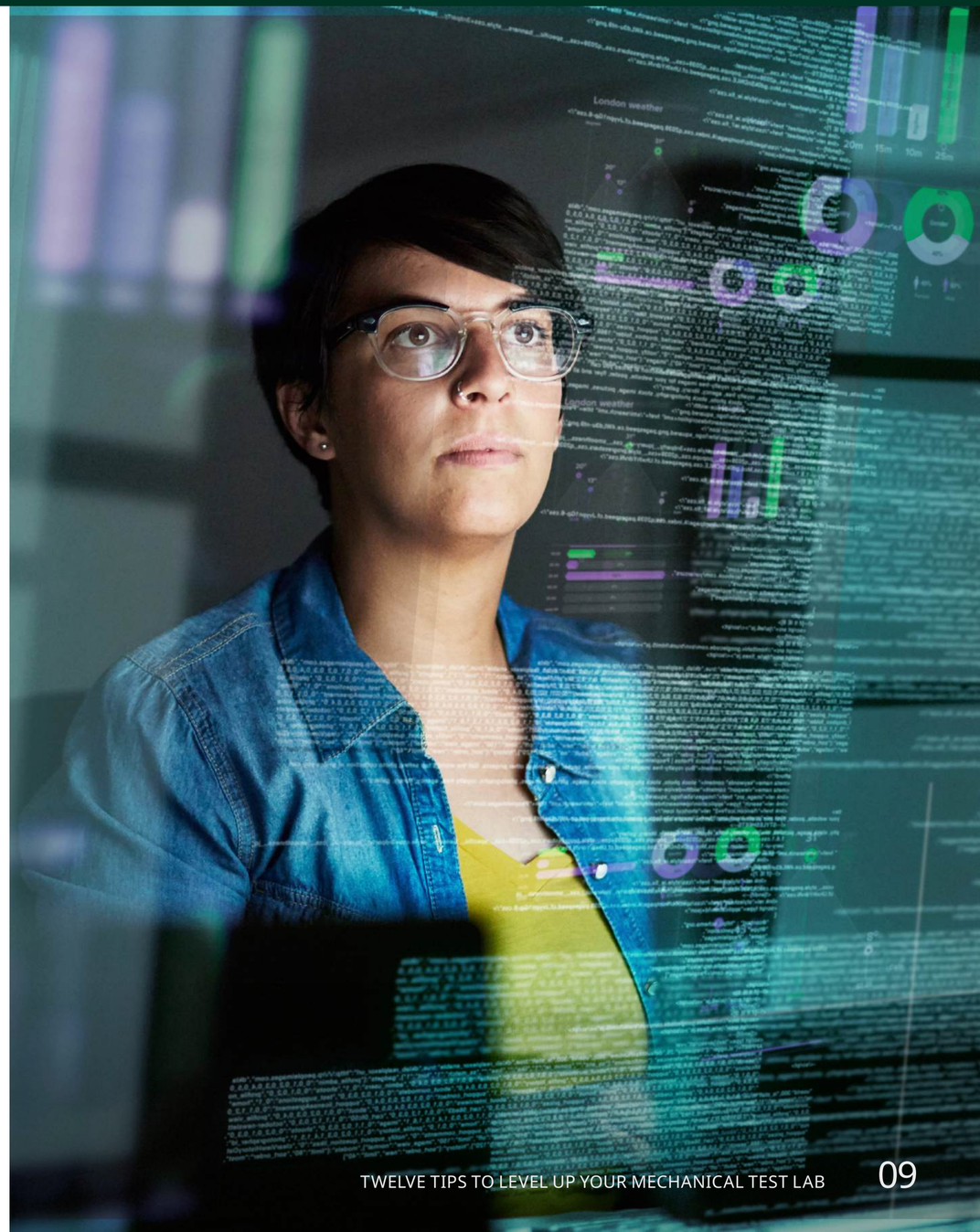
03

Control instruments, power supplies, and temp chambers to reduce operator effort and eliminate improper settings between different test runs.

04

Automatically log test system information as contextual metadata for data consumers and test system maintainers.

Culture shifts are top-down and take time. In the short term, create working groups or tiger teams to audit your use of automation. Offer rewards for automation efficiency gains. (Beer and pizza were strong motivators for young NI engineers 20 years ago...)



TIP #8

Understand the Importance of Good UI Design

If you've said, "I want a beautiful/awesome/modern interface," here are some tips:

01

Find your corporate color palette from the brand group (maybe a PowerPoint deck) to leverage both your brand identity and the color palette expertise.

02

Look for the latest UI trends in design forums. This can include iconography, animations, menu placement, fonts, and so forth.

03

Learn from other software that's not related to the test and measurement industry.

04

Consider contracting a UI designer to help.

As you look at your test system as a product, consider:

01

Interviewing test system users to understand how they like to work.

02

Screening every element to understand what users will do with the onscreen data/control.

03

Balancing your investment in your UI with its level of use. If fewer people see it and it's not used much, don't spend a lot of time on it.

04

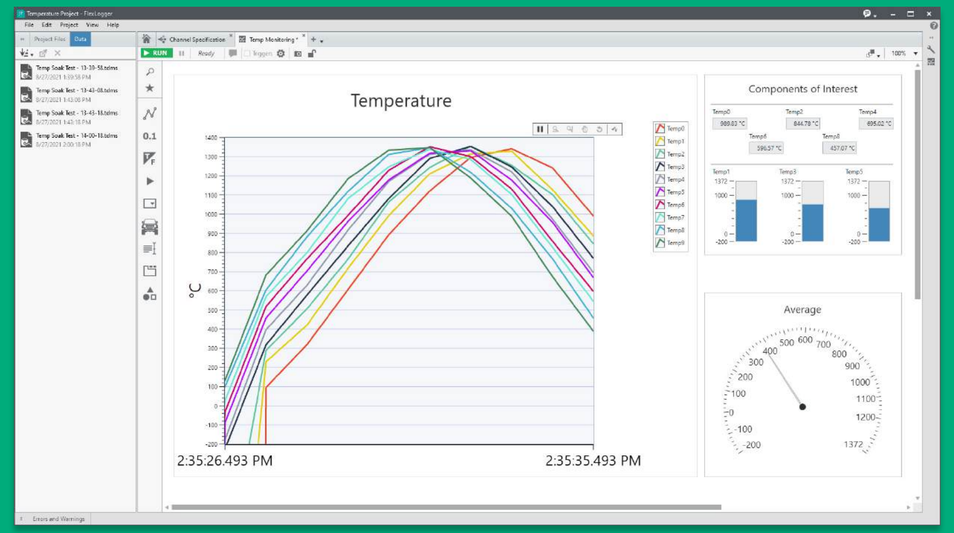
Training—is the system intuitive and need little to no documentation, or will it be intricate and require a training manual or video?

05

Standardizing user interface elements across multiple test stations to reduce training. This aligns with the modular code recommendation.

06

Keeping users comfortable—an "ugly" user interface may be helpful when you are replacing a legacy system and want to minimize change. It may be ugly—but it's a familiar ugly.



TIP #9

Balance In-House and Contractor Work

Working with contractors and systems integrators offers several advantages such as:

- **Bringing special experience, knowledge, and competencies to your project that don’t exist on your team**
- **Smoothing out workload fluctuations by helping during busy times and bridging gaps when the team is short-staffed**
- **Scaling your workforce to meet urgent deadlines**

If working with contractors is part of your strategy, consider building relationships before urgent needs happen. This will decrease onboarding time to projects when deadlines loom.



TIP #10

Have a Calibration Plan

Measurement, reporting, and traceability needs impact maintenance planning. They range from “not needed” to fully documented calibration with tracking for regulated industries like life sciences, automotive, and aerospace.

There are multiple approaches to calibration to consider. Do you shut down all test for a week of calibration, do you stagger taking systems offline for calibration and have gaps in test capability, or do you purchase spare hardware to maintain test station uptime during calibration procedures?



TIP #11

Don’t Overlook Hardware Mounting

Hardware needs space. Consider these use cases and possible solutions.

Use Case	Possible Solution
Benchtop Tests	Modern USB DAQ equipment is compact.
	Consider 19" rack solutions for benchtops when you need multiple devices like power supplies, DMMs, etc.
	DIN-rail desktop stands can keep equipment off the table to free up space.
Mobile Test Stations	Rolling cabinets/racks.
Portable Measurement Systems	Build DAQ systems into a protective case for ruggedized, portable, easier-to-use field test kits.
Test Chambers	Look at industrial enclosures or wall-mount options. Consider DIN-rail, universal panel mounts, and leaving space for future expansion with larger hardware.
Sensor Mounts, Fixtures for DUTs, Equipment Mounts	Many sensors have mounting solutions as accessories. Consider 3D printing for custom mounting solutions.

Proximity matters. Mounting your DAQ system close to your DUT requires shorter sensor wires—saving time and equipment cost while improving measurement quality. Ethernet systems installed in rugged industrial enclosures, or with rugged specs, add cost to the hardware components but can reduce overall project cost and improve measurement data.



TIP #12

Create a Cable and Connector Workbench

Cables and connectors are often an afterthought, but cable failure can lead to downtime, inefficient work, and increased system maintenance. Here are some recommendations for connectors:

- Try to standardize on fewer connector types to reduce required tools and training.
- High-density connectors are cost-effective and space-efficient but can be more difficult to work with.
- Ad hoc tests have frequent changes to sensors and channels. Consider rugged individual connectors for each channel. They will be easier to use and last longer without breaking or requiring maintenance.
- Keep a stocked workbench with:
  - Multiple spools of wire (mixed gauges and colors).
  - Solder station.
  - Heat gun and heat-shrink tubing.
  - Zip ties, cable wraps, and cable routing/retention accessories.
  - Splicers and connectors—butt splicers with solder are time savers. WAGO makes lever-actuated wire splices that are handy, especially for test system prototyping.
  - Ferrules, strippers, and crimpers. Stranded signal wires are easier to work with when crimped with ferrules.



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Our Hardware Is Modular

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The NI Ecosystem Is Open

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Why NI for Test

Every company says they save you time and money. Here are the challenges NI solves to do it.

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Changing designs, market needs, and supply chains challenge product development. Use NI tools to adapt and stay on schedule.

Standardization

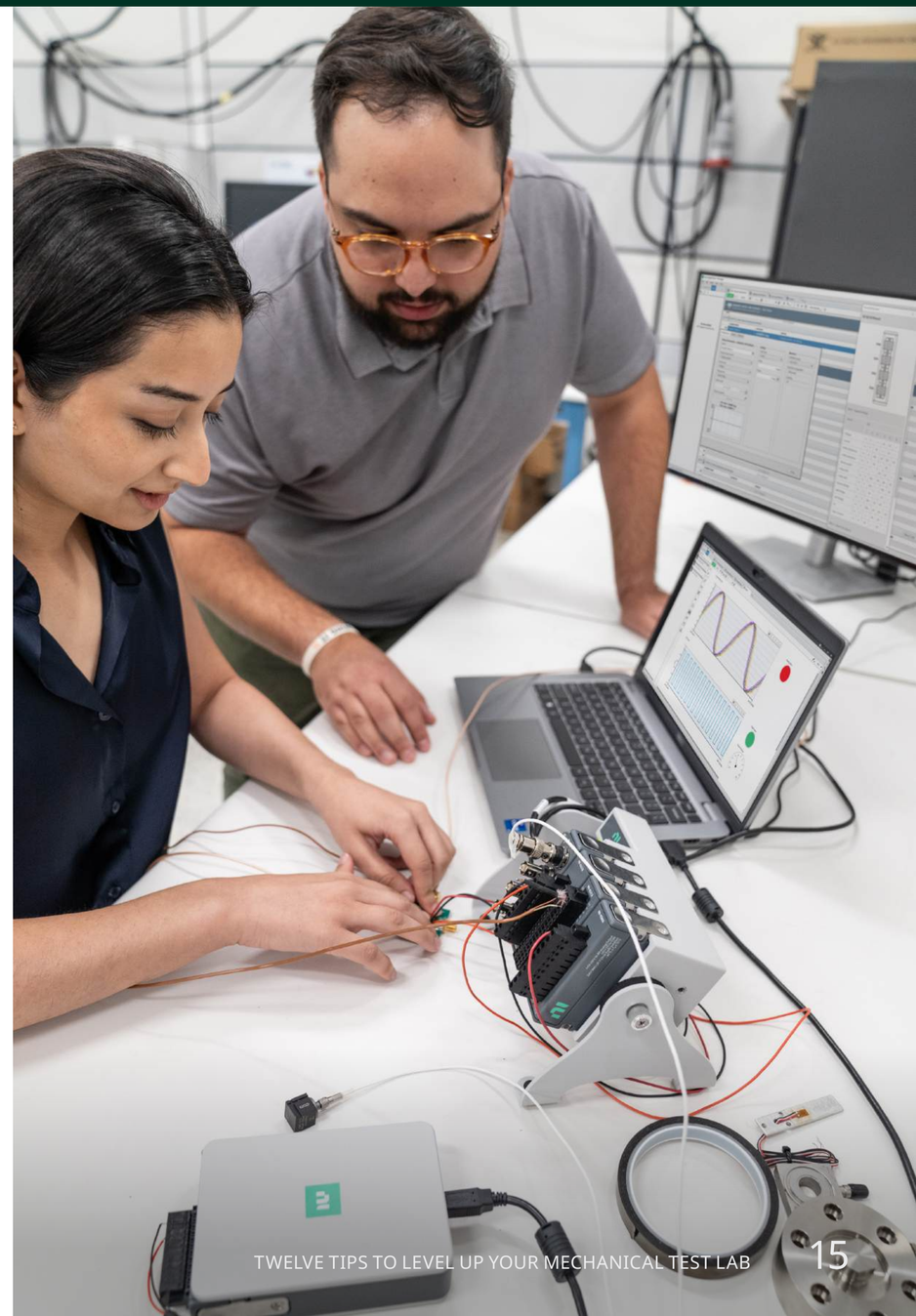
Don't spend time and money repeating work. Lower your cost of test by sharing code libraries and hardware architectures among all of your test teams.

Quality

NI is known for data throughput, acquisition rates, synchronization, and measurement quality. Better products need better test.

Productivity

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